Bridging the Gap: AI-Powered Sign Language Interpretation and Communication

line 1: 1st Joshua Djiu
line 2: Computer Science Department,
School of Computer Science
line 3: *Binus University*
line 4: Jakarta, Indonesia
line 5: joshua.djiu@binus.ac.id

line 1: 2nd Sulistyo Agung Triutomo
line 2: Computer Science Department,
School of Computer Science
line 3: *Binus University*
line 4: Jakarta, Indonesia
line 5: sulistyo.triutomo@binus.ac.id

line 1: 3rd Jeklin Harefa
line 2: Computer Science Department,
School of Computer Science
line 3: *Binus University*
line 4: Jakarta, Indonesia
line 5: jeklin.harefa001@binus.ac.id

line 1: 4th Kenny Jingga
line 2: Computer Science Department,
School of Computer Science
line 3: *Binus University*
line 4: Jakarta, Indonesia
line 5: kenny.jingga@binus.ac.id

***Abstract*—**Sign language is a basic language or communication for everyone who needs it. Often, the solutions currently available are not very easy to interpret the subtle expressions and gestures that are one of the characteristic features of sign language itself, thus possibly resulting in inaccurate translations.

This research aims to provide information on the need for innovation in sign language interpretation technology to help make it easier for deaf communities to face significant challenges and achieve meaningful progress. The research process has the task of exploring challenges by developing advances in AI technology in translating sign language. Technology with sign language translation with accuracy and effectiveness can provide solutions for the participation of deaf people in various aspects of life.

The approach in this research to technology for translating sign language focuses on developing efficiency and accuracy in translation, as well as the use of data through available sign language translation technolo

Keywords—Sign Language, Sign Language Interpretation, Artificial Intelligence, Deaf Community.

I. INTRODUCTION

Sign language is a language or basic communication tool for deaf people throughout the world in facing challenges with limited availability in obtaining quality sign language translations. Each of these challenges hinders or hinders the full support and integration of people with hearing loss in various perspectives or aspects of life such as work, social intuition, etc. The purpose of sign language translation can be, for example, to convert written language into a sign video or to extract related sentences from a spoken language video in which a person repeatedly signs [2].

Luckily, Artificial Intelligence (AI) offers promising arrangements. AI-powered sign language interpretation programs attempt to bridge this creation by deciphering sign language into spoken language in real-time. Clients can be associated with the program consistently through hand developments and facial expressions, and the program at that point produces comparable content or discourse [14].

In processes or process flows through innovation that have shown significant progress, this research identifies the challenges available, as well as improvements in artificial intelligence (AI) assistance used for sign language translation. This process could take the form of an analysis of the potential for revolutionizing communication needs for deaf or hard of hearing communities. Even though the process flow is still in the development process, this innovation process can show basic and important progress. Therefore, with the availability of improvements, as well as known challenges, it will be next to identify the transformative impact of artificial intelligence (AI) on increasing inclusivity in communication by deaf or hard of hearing communities [14].

1.1 Problem

Significant barriers or alignments in the development of sign language translation technology that hinder full integration, as well as communication accessibility for deaf or hard of hearing communities. Currently, the availability of solutions also often results in difficulties in interpreting things such as expressions or movements that are characteristic of the required language, so this will result in less accurate translators. These obstacles can also come from a lack of innovative approaches to the development of sign language translation.

This important gap or obstacle really requires innovation or innovation of iconic language translation technology to achieve progress and improvement. Therefore, by developing advanced algorithm processes such as the use of deep learning processes, as well as artificial intelligence (AI) will provide more effective and objective sign language translation solutions.

1. 2 Objective

Sign language translation has the aim or purpose of developing sign language translation technology that is more accurate, structured and effective through developing technology in providing communication opportunities for deaf people or people who are hard of hearing, so that with this opportunity, deaf people or people who People with hearing loss can participate in and participate in various social interactions.

Sign language translation technology also has other goals, for example in the form of innovative use of sophisticated machine learning algorithm technology or what is usually referred to as machine learning (ML), as well as the development of pattern recognition to more accurately receive and encourage sign language nuances and connections with each other [4].

1.3 Benefit

The advantage of sign language translation technology with artificial intelligence (AI) is that it allows people with hearing disabilities to have access to better health services, thereby preventing transmission or health problems that are often experienced by communities with hearing or sign language impairments. . Therefore, society will better understand their needs

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and rights, if there is a deep awareness of the resource challenges experienced by deaf or hard of hearing people.

The development of a broader technological ecosystem is also one that includes native language translation technology which provides, encourages and develops this through the process of creating communication platforms specifically designed to support communities who experience hearing loss or hearing disabilities.

II. LITERATURE REVIEW

The research papers contained in this dynamic literature have the aim of creating an introduction using ontology (deep learning) which is used as a cloud computing environment and developing technology or systems from DGDTS which function as dynamic semantic translation. This ontology system with video classification has 2 models, namely the ConvLSTM and 3DCNN models which are used in extracting spatio temporal features in movement [1].

Epoch	3DCNN+ConvLSTM		
	Our dataset	One-signer LSA	LSA40
10	81.8	92.9	98
20	85.4	97.6	98
30	90.9	97.6	98
40	87.3	95.3	98.2
50	92.7	96	98.2
60	85.4	96.8	98.2
70	87.2	96.8	98.2
80	89	96.8	98.2
90	93.7	99.2	98.5
100	94.5	99.2	98.5

Example of a dataset containing epoch variations

The explanation in the image above is the process of summarizing accuracy with the architecture of the ConvLSTM and 3DCNN models. If the process uses a small set, there will definitely be changes in the accuracy of the number of iterations (better accuracy will still be obtained even if there is a largest epoch). This research conducted model training for 100 periods to obtain the best detection accuracy using 3 available data sets [1].

Dataset	No. signer	No. classes	No. testing videos	No. training videos	Classification	3DCNN + ConvLSTM Recogni	Execution time on windows	Execution time on cloud	3DCNN Recognition accuracy
LSA [24]	10	40	400	1600	%100	98.5%	14 h	2 h	97.7%
LSA one signer	1	64	128	192	%100	99.2%	2 h	9 min	96.8%
Our recorded gestures	1	11	55	220	%100	94.5%	2 h	9 min	85.4%

Example of 3 datasets containing 100 epochs

The explanation of the 3 datasets of 100 epochs contained in the image above provides information on the test results on these 3 datasets, that changes or progress are shown from model comparisons between ConvLSTM and 3DCNN and assisted by the Windows Anaconda Neighborhood tool in the runtime training process [1].

The research papers contained in this literature create a strategy or plan for the introduction of Indian Sign Language or what is abbreviated as ISL, where this strategy recommends GRU and LSTM for hand signal data or datasets with aims such as getting more accurate words general (for example the words "good morning", "good evening", etc.) [11].

In technology, the translation method must be based on detection, which is also included as the main step in the system or Sign Language Transformers. This technology is a transformer-based system that is used to read or understand sign language translation recognition [2].

The research papers contained in this literature have 2 approaches, namely [3]:

- The results of a qualitative approach are by obtaining several models of qualitative results themselves. the performance process is by comparing the translation between the reference text and MLSLT which will provide a translation from the language itself to another language, and vice versa,
- Meanwhile, another approach is to experiment with the performance process using two benchmarks, namely:
- MLSLT training model to better recognize the model as a BSLT benchmark,
- The original training model was used to train the model as a benchmark for MSLT.

An approach that uses identification of the characteristics of sign language translation. The characteristics of sign language translation or what is abbreviated as SLT is the process of converting sign language into spoken language [4]. There is also something other than the previous approach, namely a glove-based approach with performance results that can be said to be very accurate, but it will not look comfortable when gloves are worn in real world conditions every day. Meanwhile, vision is dynamic and statistical with performance processes or goals in sign language in the form of words or finger spelling [12].

Automatic sign language recognition or translator technology is proposed with the aim of making it easier to provide better communication tools or systems for deaf or non-hearing-impaired communities [5], and overcoming challenges such as not everyone having knowledge of understanding language. signals, so they can have a small impact on communication with each other [5].

This research paper has a method or approach that can be said to be effective in obtaining detailed hand shape images or patterns in ASL symbols [13]. There are several advantages of this approach, namely as follows [13]:

- Has the potential in real time and quickly,
- Has the advantage of saving costs,
- Good accessibility improvements,
- A promising system in the field of character recognition,
- Able to recognize characters written in the air,
- Improvements in better accuracy.

The development of sign language recognition technology uses a deep learning model or commonly referred to as deep learning which will be analyzed into precise sensor data and signals. then, signs or symbols that were previously known will be translated into text or also audio, making it easier for the community of people with hearing disabilities and for

communities who do not experience hearing impairment to communicate without any obstacles [14]. Previous technological developments could be related to human vision-based SLR systems in a deeper or wider range of languages [15].

The research papers contained in this literature review provide a model or method that can be said to be cost-effective through the application of VH access, where the model is based on the development of virtual person motion acquisition technology that can support a broad user base [6]. The development of technology with other models such as the Ham-NoSys English text translator or generally referred to as PSL also has complex performance in the form of producing numbers or letters which will then be converted into SiGML or XML [7].

The way the CNN system works or performs in this research is by expanding it to other sign languages by creating an appropriate training model. If the sign language translator system spells or recognizes finger movements accurately, for example 90% or more, it cannot be further processed [8].

This research paper identifies two types of process models which are as follows [9]:

- The design training process is carried out by cross validating the model with the processed hybrid,
- The process of evaluating the capabilities of Monte Carlo through understood data (ASL) is carried out by only focusing on obtaining random samples to obtain numerical results.

This project or research has challenges, for example when the technology or system will be processed at a later stage with the requirement that one of them be multilingual, where availability for various languages is inconsistent which is one of the obstacles in system processing [10].

This research has a type of model called YOLO whose function is to obtain precision and accuracy information, as well as positive correlations observed during the training period. There are also other models which, if combined, will become technological advantages in their own right, namely the LSTM model based on a framework process which is better for the continuous character classification process [16].

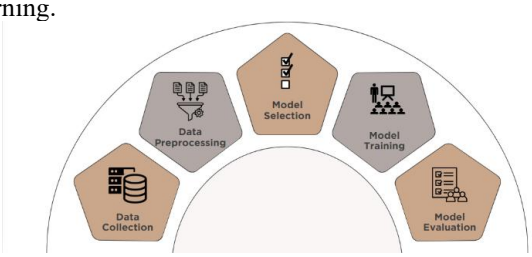
Pakistani Sign Language or abbreviated as PSL contains pre-processing processes, data collection or expanding data sets, as well as classifying various models and also real-time and fast trackers or detectors [17]. Other research papers include a survey in sign language which is used to search for available material by developing its quality, as well as the process of classifying data sets or datasets [18].

Research from this literature discusses the use of models that have been trained in various fields, for example detection models and sign language recognition models originating from Bengali. The potential for deep learning methods or models or commonly referred to as deep learning apart from the two previous models can also provide significant improvements and benefits to sign language recognition communities [19].

The benefits of sign language translation technology can provide a comfortable medical experience through accuracy and ease of use. This technology or system will also help in advancing real-world and future work that will focus on expanding the performance of sign language words or spelling, automatic data collection, etc. [20].

III. WORKFLOW

The image below is the process used in sign language translation tests using artificial intelligence or machine learning.



Sign language translation experimental process

IV. METHODOLOGY

This sign language translation technology has several approaches, one of which is an approach that focuses on developing technology to increase efficiency and perfect the accuracy of sign language translation. In this approach, there is a need to link the use of artificial intelligence (AI) and machine learning (ML) to develop translation technology that is more flexible and adaptive. This sign language translation technology also has another approach in the form of a database approach that is used on data from the sign language itself by training and testing models for the sign language translation. A data-based approach also requires a process of analyzing data from various sign languages in order to develop a translation system that is able to recognize sign language patterns well. Likewise, other approaches such as testing, or evaluation approaches require linkages in the evaluation and testing of the translation system with the aim of verifying its effectiveness. Another related testing and evaluation approach could be testing the performance or work process of the system in a communication situation. a community-engaged approach will also actively connect the hearing disability community in the process of developing sign language translation technology with the goal of collaboratively gathering input and developing solutions that meet their availability.

The testing or evaluation approach with sign language translation technology can also be interconnected with data collection through the initial stages until the end, such as the initial phase, namely collecting sign language data extensively (this data can be data in which there are video recordings of native sign language speakers whose contents express various words, phrases, or sentences in the target sign language). Furthermore, it involves adding metadata to each video recording (for example a transcription of sign language text that corresponds to the video expression provided) or is generally referred to as data annotation. then, the data is processed first to be used in training the sign language translation model (such as data normalization, sign segmentation, etc.). Data annotation is used to train a model

with the aim of more accurately understanding patterns or movements in sign language, which will then be evaluated to measure its potential effectiveness using separate testing data.

4.1 Data or Datasets



Overview of data or dataset

This part can be said to be the process of collecting data by developing and creating your own dataset to better ensure the accuracy of the data in accordance with the availability in this research. The complex process can include taking photos of the hand in various patterns, as well as movements by marking dots or what are known as landmarks on the hand itself. The data or dataset obtained includes 27 different classes (each class will be recorded by unique hand patterns, positions and movements) and includes 100 images for each class until the data collection has a total of 2,700 images. The data or dataset collection process is carried out manually, to ensure the image quality is clearer in the marked landmarks. Therefore, collecting data or datasets manually or yourself can control variations in lighting sensors and other conditions that may make changes to the results according to requirements.

4.2 Data Preprocessing

This data processing includes the initial step of obtaining and collecting examples of hand signals from ASL. then this data is processed using the MediaPipe tool, as well as OpenCV which aims to capture landmarks in hand. Next, the obtained landmarks are used as input for training a classification model with the Scikit-Learn tool or library and the random forest algorithm. As a result, the training model tested by testing this data achieved 100% accuracy, which marks the effectiveness of the method or training in the process with the training model and model testing using training data or datasets, as well as testing data or datasets.

4.3 Model Selection

The choice of model used is a random forest classifier model with learning algorithm performance that combines several decision trees with the aim of forming a more accurate, broad or complex model. Then this random forest will work by combining predictions from several decision trees individually which is useful for updating and increasing accuracy. This learning algorithm recognizes relationship patterns between fingerprints (signature class relationships), the results also achieve 100% accuracy in classifying hand signs in sign language which provides a potential sign in capturing and recognizing signatures with high accuracy.

4.4 Model Training

The drilled model has three steps in the machine learning (ML) process. First, prepare the data by taking a picture of the hand using a tool such as a webcam, and creating hand landmarks using the MediaPipe tool. second, using the random forest classifier model in the scikit-learn library, which works by taking fingerprint data which will be used as input in learning the model by detecting patterns or relationships between signatures and fingerprints. Third, test the model by testing unprocessed data or datasets.

4.5 Model Evaluation

The model being evaluated will be carried out by separately testing the data contained in the training data. Then, the previously evaluated model will be tested in predicting classes from test data, as well as a comparison between the prediction results and the original or actual labels. In the accuracy section, it can be seen, when the evaluation measure used will show the ability or how well the model is predicting test class data.

V. RESULTS AND DISCUSSION

The results of this research use computer vision, as well as deep learning or machine learning (ML) in developing native language translation technology to capture or detect and understand native language hand signs with a webcam. This sign language translation detection tool or device can classify three sign language signatures. This detection or classification uses manual measurements or comparisons from the MediaPipe library, as well as the random forest classifier model in Scikit-learn. From the workflow or previous performance, the result is a device that has been developed that reaches 100% accuracy, which provides a sign or information that the potential of the model in the process of classifying sign language signatures is accurate or precise. The function of this technology or device is not only devoted to making it easier for communities of people with hearing disabilities or for people who are hearing impaired who communicate in sign language using a webcam, but can also make it easier to learn sign language in an adaptive and interactive manner.



Overview using finger pose for the letter 'A'



Overview using finger pose for the letter "I LOVE U"

Explanation of the steps for finger poses in the image above, if you put your finger on the 'A' in ASL sign language, you can make a fist with one hand, making sure the thumb doesn't go into the fist (holds up). Then keep the other fingers, including the index, middle, ring and little fingers, hooked inwards. At the same time, in ASL sign language, you can raise one hand and open the fingers, then fold the middle and ring finger down, keeping the thumb, index finger and little finger crossed for the "I LOVE U" finger position. ASL letters can also be combined with this form. For example, if you raise your little finger to make the letter 'I', your thumb and index finger will point to the letter 'L', and raising your thumb and little finger will point to the letter 'U'.

The technology or device developed to detect sign language makes it possible to consider accessibility, because it has a user interface that makes it possible to determine the navigation process on the technology easily. The technology or device developed by this sign language detector can also support real-time and fast feedback so that users can easily find out directly the results of improvements in sign language.

VI. CONCLUSION

This research aims to develop a native language translator with computer vision and deep learning or machine learning (ML) which functions in understanding and recognizing native language hand signs with the availability of a webcam. The accuracy obtained is 100% which is supported by special data collection containing 2,700 images from 27 classes, and uniquely representing hand patterns, positions or movements. Data processing includes manual landmark detection using the OpenCV tool, as well as MediaPipe which is carried out by training a random forest classifier model. This model is said to be very adaptive and effective in real-time systems because of its potential in handling several different data. The purpose of training and testing or evaluation of the model is carried out with the aim of ensuring that the classification has good accuracy and will also provide a sign or information about the potential or ability of recognizing sign language translation.

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